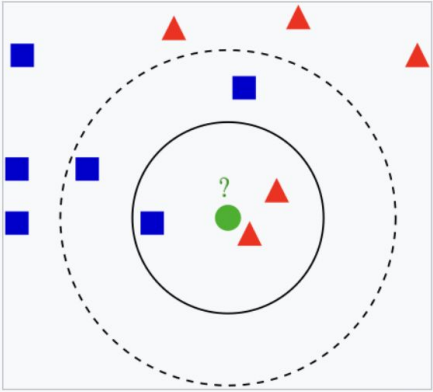
Our first ML Model

- k-Nearest neighbors (k-NN)
  - Supervised learning
  - Non parametric ***(Distance based method)***
  - Both for Classification and Regression solutions

This is a red triangle ▲ *vs blue square* ■ *classification problem*



Example of *k*-NN classification. The test sample (green dot) should be classified either to blue squares or to red triangles. If *k = 3* (solid line circle) it is assigned to the red triangles because there are 2 triangles and only 1 square inside the inner circle. If *k = 5* (dashed line circle) it is assigned to the blue squares (3 squares vs. 2 triangles inside the outer circle).

src: https://en.wikipedia.org/wiki/K-nearest_neighbors_algorithm

Quiz!!!

- You have two (2) **ML engineer** friends with their **years of experience** and **salaries** are as follows

	profession	years-of-experience	salary
0	ML engineer	2	120000
1	ML engineer	5	160000

Quiz!!!

- You have two (2) **ML engineer** friends with their **years of experience** and **salaries** are as follows
- What you expect the salary would be for two other ML engineers
  - With no experience
  - With 3 years of experience?

	profession	years-of-experience	salary
0	ML engineer	2	120000
1	ML engineer	5	160000
2	ML engineer	0	?
3	ML engineer	3	?

Quiz!!!

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CIS 635 Knowledge Discovery & Data Mining

ML Introduction: Linear Regression (part 1)

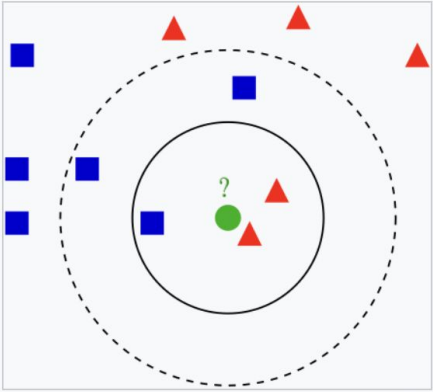
What we'd like to accomplish today

- General Concepts: ***Straight Line** to **Linear Regression***
- Gradient Descent Algorithm
  - ➜ A simple two parameter **Linear Regression** model
  - ➜ Hands on **Notebook implementation**
- QA

Our first ML Model

- k-Nearest neighbors (k-NN)
  - Supervised learning
  - Non parametric ***(Distance based method)***
  - Both for Classification and Regression solutions

This is a red triangle ▲ *vs blue square* ■ *classification problem*



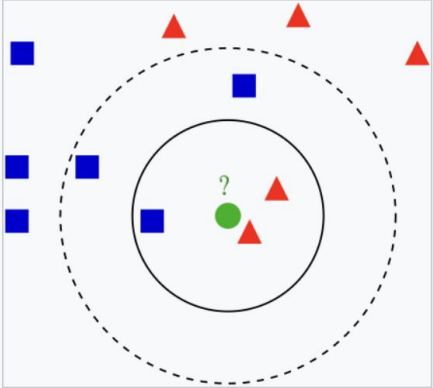
Example of *k*-NN classification. The test sample (green dot) should be classified either to blue squares or to red triangles. If *k = 3* (solid line circle) it is assigned to the red triangles because there are 2 triangles and only 1 square inside the inner circle. If *k = 5* (dashed line circle) it is assigned to the blue squares (3 squares vs. 2 triangles inside the outer circle).

src: https://en.wikipedia.org/wiki/K-nearest_neighbors_algorithm

Our first ML Model

What other classification problem you can think of?

- *Email Spam filter:* $y \in \{0,1\}$
- *Face recognition:* $y \in \{$ ***you, your friend, ...***$\}$
- *Cancer detection:* $y \in \{0, 1\}$
- *News topic detection:* $y \in \{$ ***politics, sports, ...***$\}$
- *Sentiment classification:* $y \in$ ***{happy, sad, ...}***



Example of *k*-NN classification. The test sample (green dot) should be classified either to blue squares or to red triangles. If *k = 3* (solid line circle) it is assigned to the red triangles because there are 2 triangles and only 1 square inside the inner circle. If *k = 5* (dashed line circle) it is assigned to the blue squares (3 squares vs. 2 triangles inside the outer circle).

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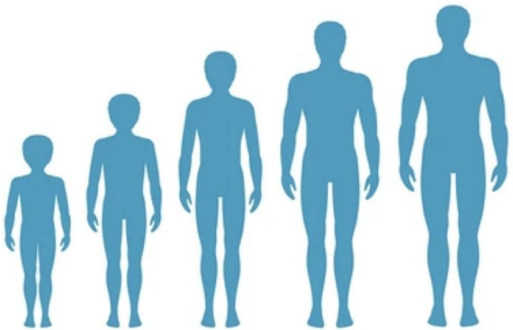
We will be learning today about Regression

- *Insurance cost : $y \in R$*
- *House price: $y \in R$*
- *Weather prediction: $\boldsymbol{Y \in R}$*
- *Energy consumption: $y \in R$*
- *Sales forecasting: $y \in R$*

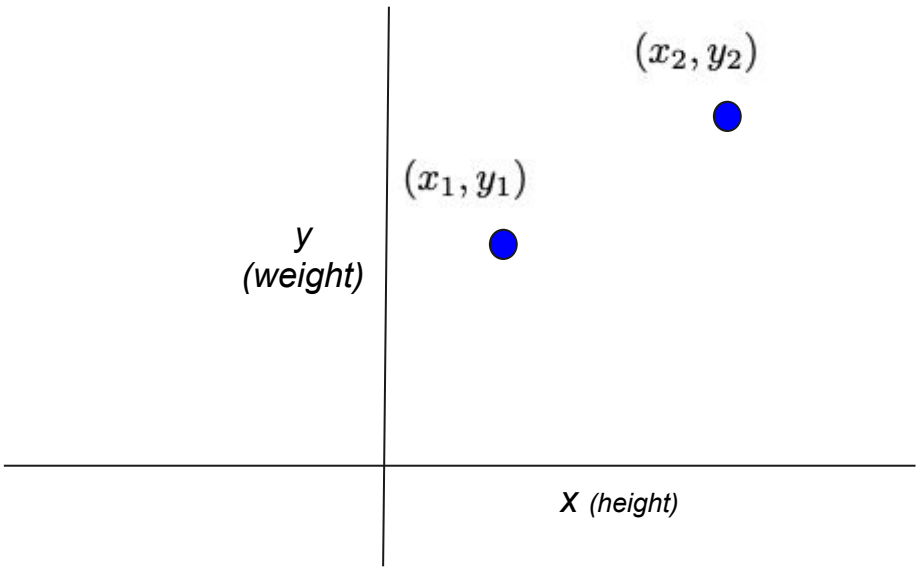
We will be learning today about Regression

- *Person's weight : $y \in R$*

$$f(y|x=height)$$

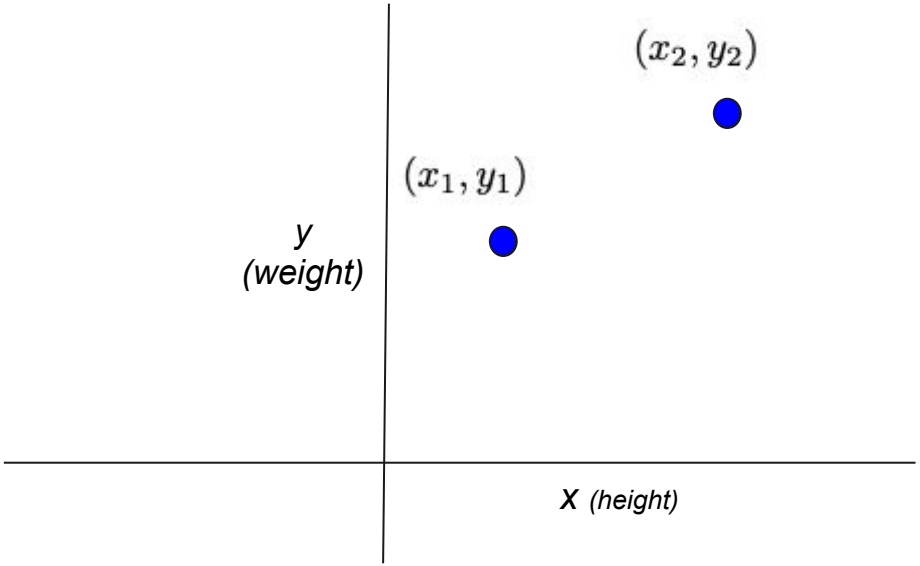


Given these two data points



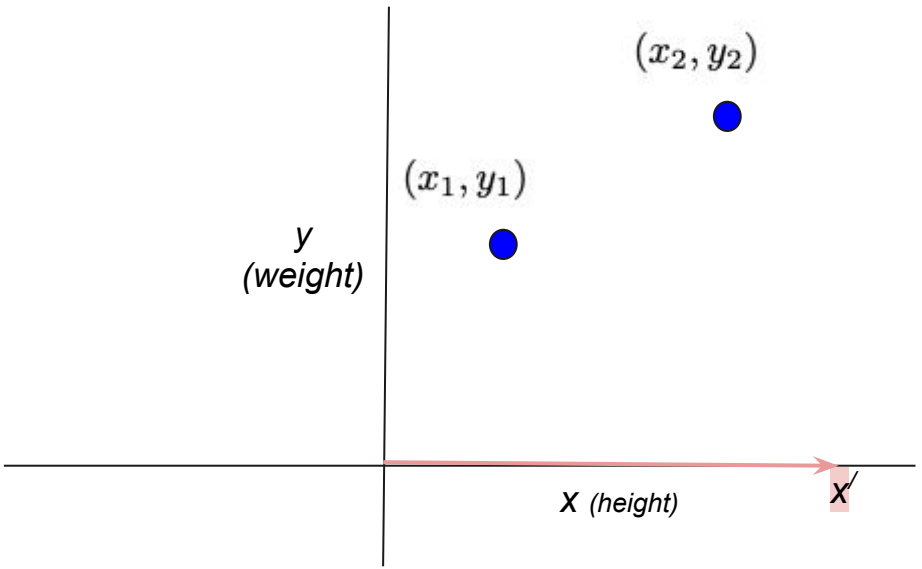
Given two known data points (x_1,y_1), and (x_2,y_2), and

k-NN Regression



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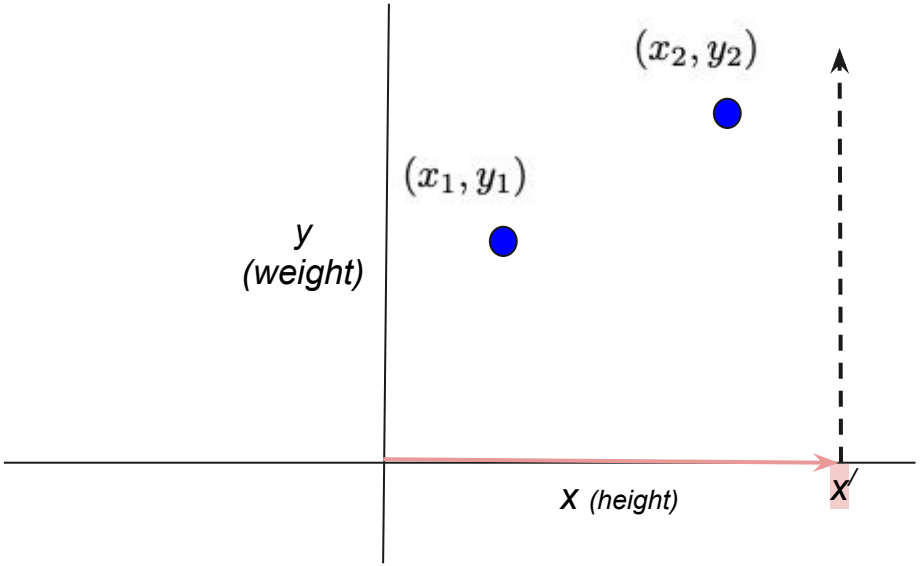
k-NN Regression



Given two known data points (x_1,y_1), and (x_2,y_2), and

- for test input $x^{/}$, you have to predict $y(x^{/})$.
- I.e. you have to plot ($x^{/}$, ?)

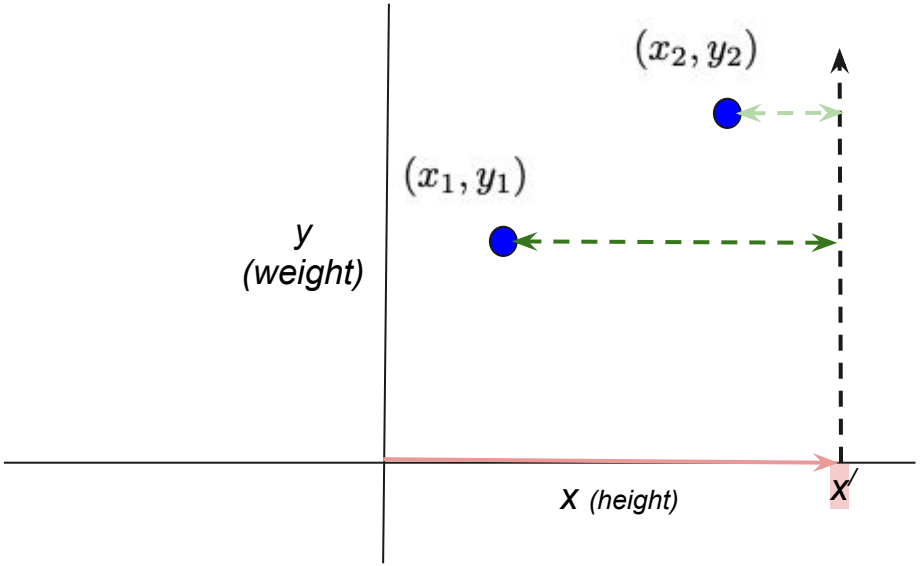
k-NN Regression



Given two known data points (x_1,y_1), and (x_2,y_2), and

- for test input x', you have to predict $y(x')$.
- I.e. you have to plot (x', ?)
- To estimate the distances let's draw the vertical line

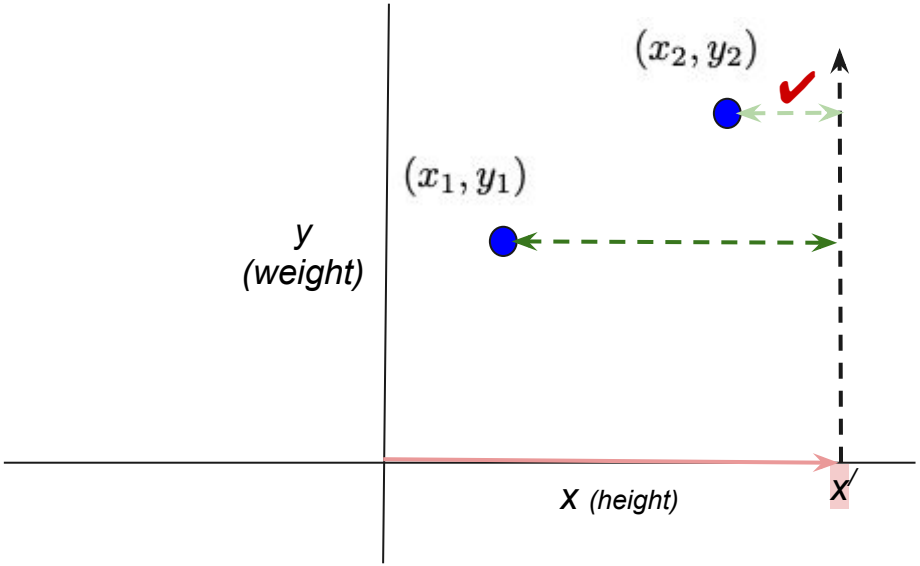
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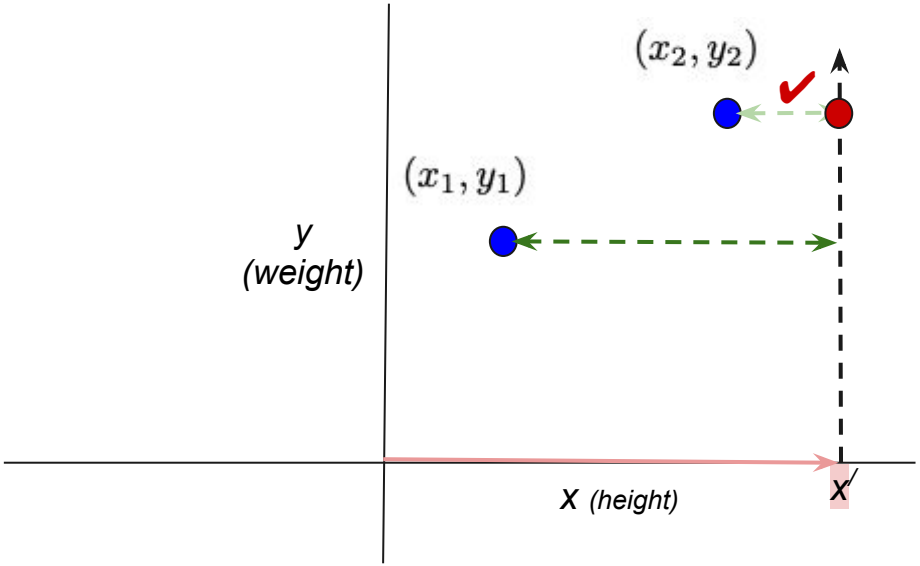
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-

k-NN Regression

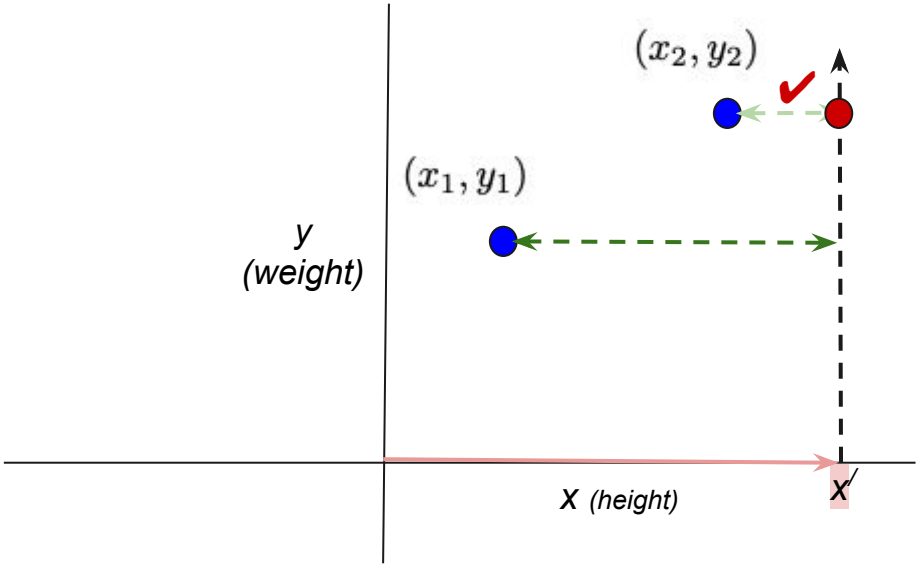


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k-NN Regression

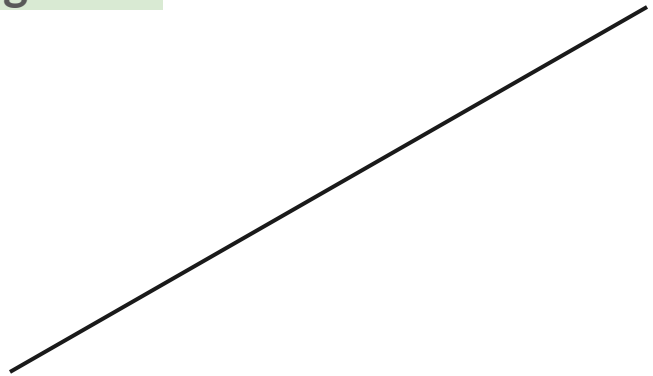




Can you think of a smarter solution?

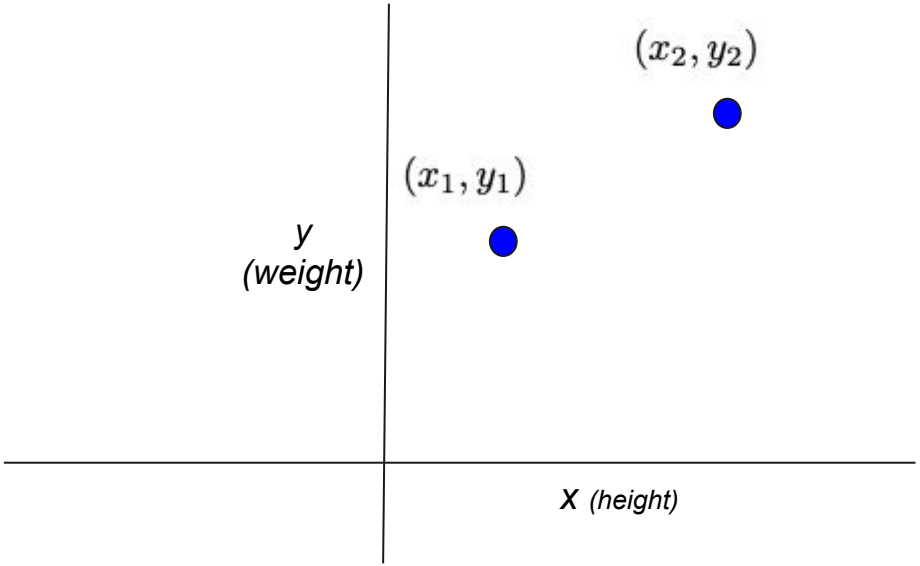
Our Second (Supervised) Model

A simple straight line



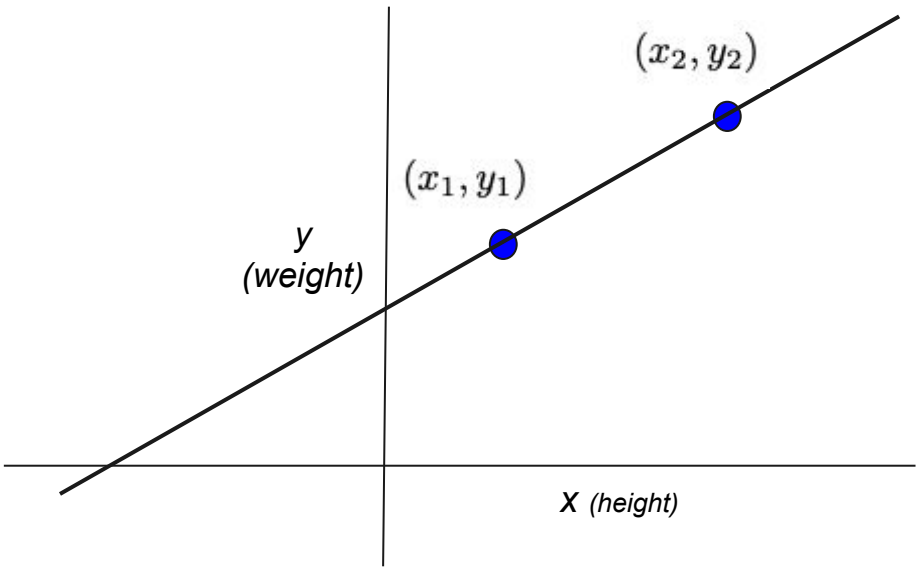


Linear equation, a quick review



Given two data points

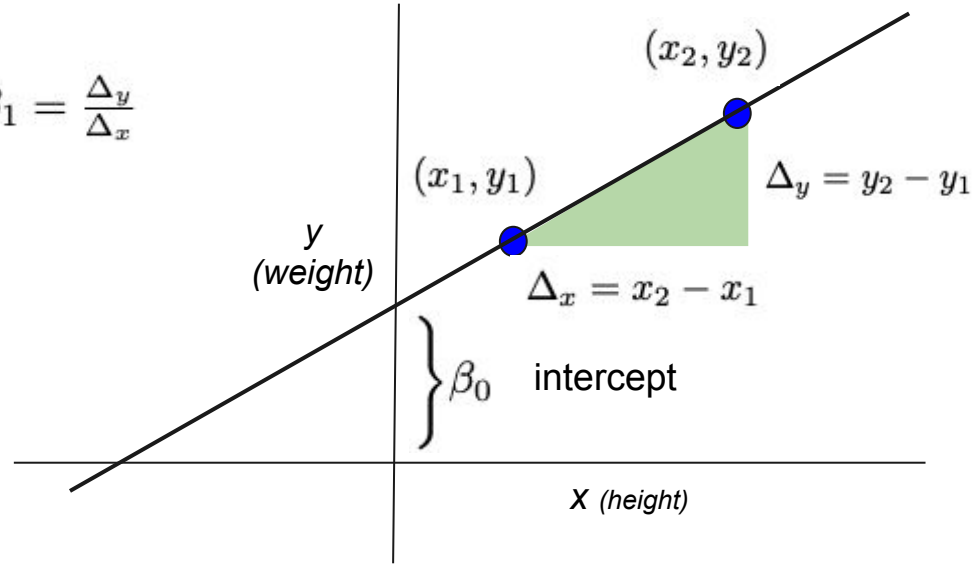
Linear equation, a quick review



We can fit a linear equation

Linear equation to a linear function, a quick review

slope $\beta_1 = \frac{\Delta_y}{\Delta_x}$



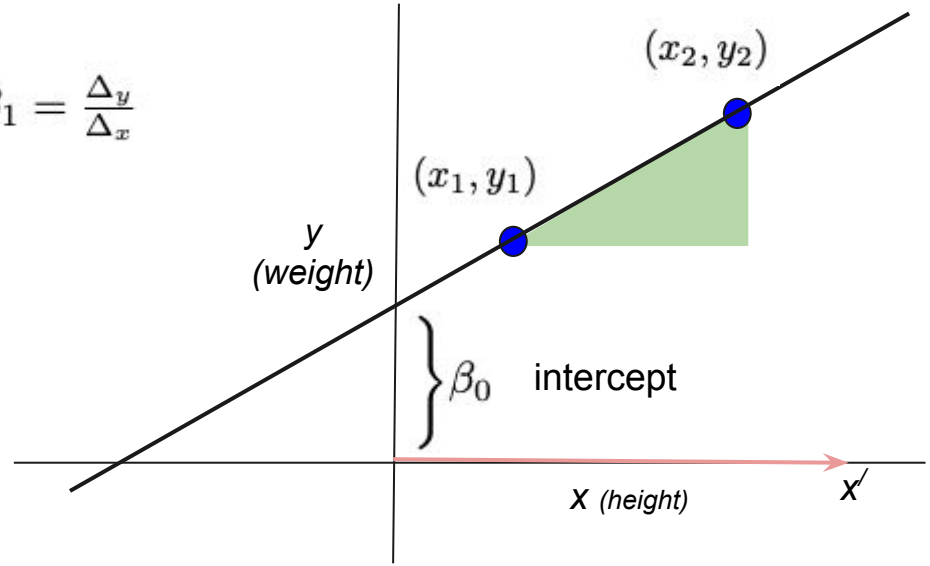
- This linear equation can be used to explain the relationship between the two axes ***x*** (independent variable) vs ***y*** (dependent variable) - as

$$y = \beta_0 + \beta_1 x$$

- A simple model with parameters: **slope**, and **intercept**
- For any given x', this model can predict $y(x')$ using the above equation.

Linear equation to a linear function, a quick review

slope $\beta_1 = \frac{\Delta_y}{\Delta_x}$



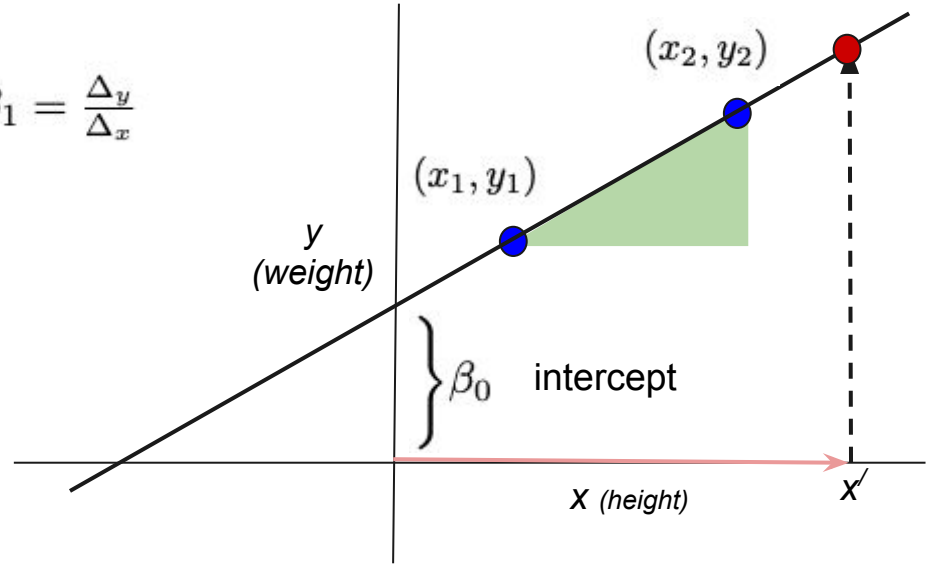
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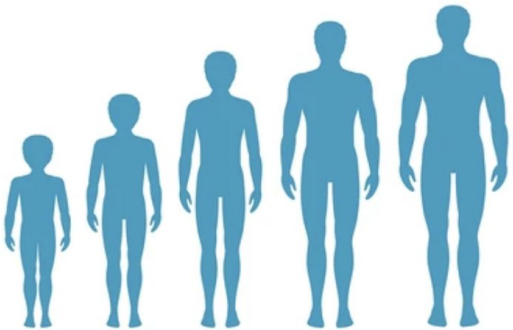
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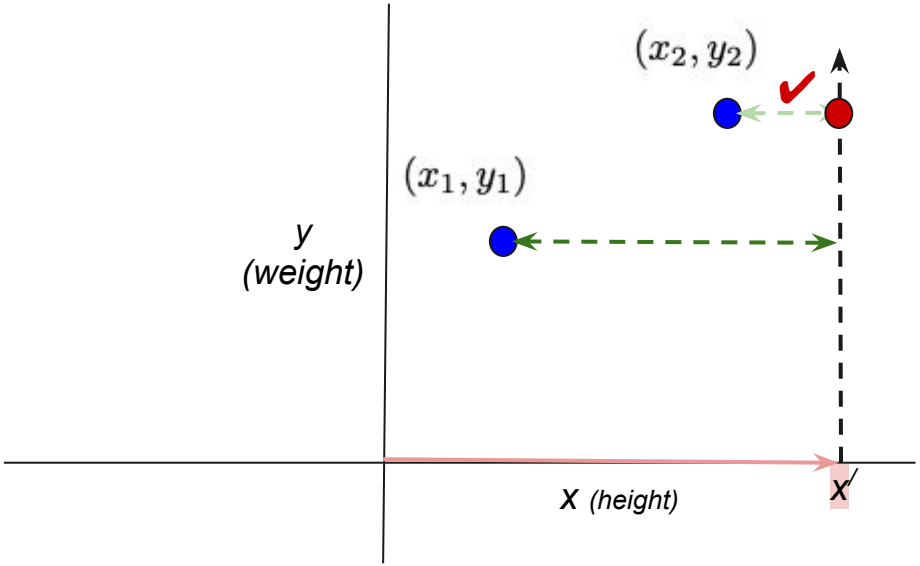
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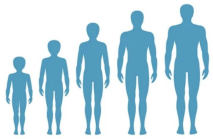
k-NN vs. Linear Equation

Model (class) comparison!



k-NN Regression





Failed to capture the LR relationship

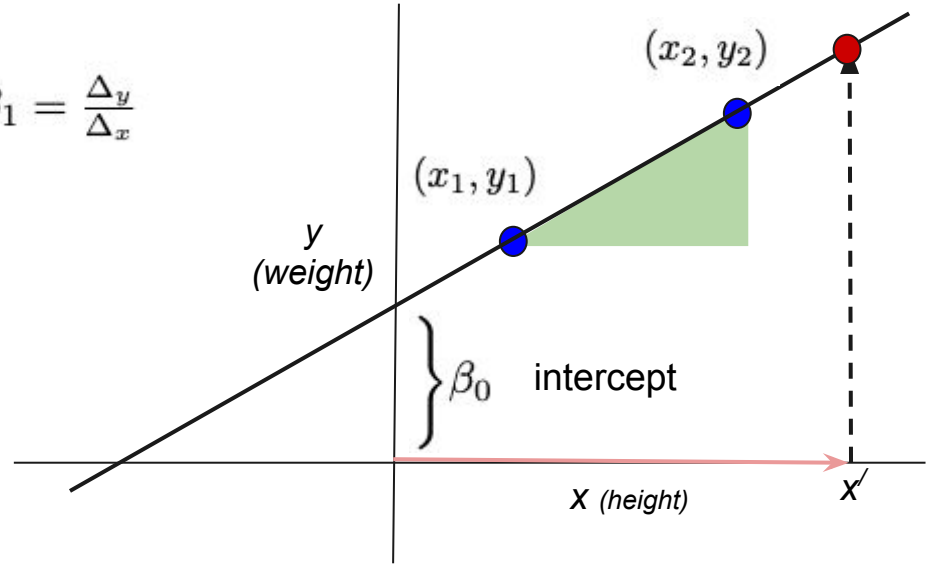
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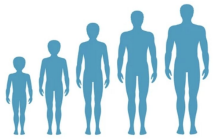
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Linear Equation

slope $\beta_1 = \frac{\Delta_y}{\Delta_x}$





Was able to capture the LR relationship

- This linear equation can be used to explain the relationship between the two axes ***x*** (independent variable) vs ***y*** (dependent variable) - as

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QA